

13 Chapter

PIPE & CISTERN

◆ TYPE - I ◆ COMPLETELY FILLING OR EMPTYING

- Two pipes A and B can fill a tank in 15 hours and 18 hours, respectively. Both pipes are opened simultaneously to fill the tank. In how many hours will the empty tank be filled?
SSC CPO 23/11/2020 (Shift-01)
(a) $7\frac{2}{11}$ (b) $9\frac{2}{11}$
(c) $10\frac{2}{11}$ (d) $8\frac{2}{11}$
- Two pipes P and Q can fill a tank in 36 minutes and 45 minutes, respectively. If both pipes are opened together, the time taken to fill the tank is:
SSC CGL 13/09/2024 (Shift-03)
(a) 81 min. (b) 20 min.
(c) 40.5 min. (d) 10 min.
- Two pipes A and B can fill a tank in 12 hours and 18 hours, respectively. Both pipes are opened simultaneously. In how much time will the empty tank be filled completely?
SSC CPO 25/11/2020 (Shift-01)
(a) 8 hours
(b) 10 hours 24 minutes
(c) 9 hours 30 minutes
(d) 7 hours 12 minutes
- Pipe L can fill a pool in 30 hours and pipe M in 45 hours. If both the pipes are opened in an empty pool, how much time will they take to fill it?
SSC CGL 10/09/2024 (Shift-02)
(a) 24 hrs. (b) 17 hrs.
(c) 18 hrs. (d) 20 hrs.
- Two pipes X and Y can fill a tank in 14 hours and 21 hours, respectively. Both pipes are opened simultaneously to fill the tank. In how many hours will the empty tank be filled?
SSC CGL 25/09/2024 (Shift-02)
(a) $8\frac{2}{5}$ hours (b) $6\frac{2}{5}$ hours
(c) $7\frac{2}{5}$ hours (d) $5\frac{2}{5}$ hours
- Pipe A can fill an empty tank in 18 hours and pipe B can fill the same empty tank in 24 hours. If both the pipes are opened simultaneously, how much time (in hours) will they take to fill the empty tank?
SSC CGL 14/07/2023 (Shift-04)
- Two pipes A and B can fill a tank in $1\frac{1}{3}$ hours and 2 hours, respectively. If both the pipe are opened simultaneously, then in how much time will the empty tank be filled?
SSC CGL 12/09/2024 (Shift-03)
(a) $1\frac{1}{4}$ hours (b) 55 minutes
(c) 48 minutes (d) $1\frac{2}{3}$ hours
- Pipe A can fill 50% of the tank in 6 hours and pipe B can completely fill the same tank in 18 hours. If both the pipes are opened at the same time, in how much time (in minutes) will the empty tank be completely filled?
SSC CGL 26/07/2023 (Shift-02)
(a) 420 (b) 425
(c) 432 (d) 435
- 5 pipes are required to fill a tank in 1 hour and 40 minutes. How long (in minutes) will it take to fill the same tank if 4 pipes of the same type are used?
SSC MTS 11/09/2023 (Shift-02)
(a) 115 (b) 140
(c) 125 (d) 150
- 15 taps can fill a tank in 36 minutes. How many taps will be required to fill the tank in one hour?
SSC CHSL 15/08/2023 (Shift-02)
(a) 12 (b) 9
(c) 8 (d) 6
- A pipe can fill an empty tank in 5 minutes and another pipe can empty it in 6 minutes. If both pipes are opened simultaneously, how long (in minutes) will it take to fill the empty tank?
SSC CGL 13/09/2024 (Shift-02)
(a) 33 min. (b) 30 min.
(c) 35 min. (d) 25 min.
- A tank can be filled by pipe A in 4 hours and pipe B in 6 hours. At 8:00 a.m., pipe A was opened. At what time will the tank be filled if pipe B is opened at 9:00 a.m.? **SSC CGL 23/09/2024 (Shift-01)**
(a) 10:16 a.m. (b) 10:22 a.m.
(c) 10:48 a.m. (d) 10:18 a.m.
- Pipes A, B and C can fill a tank in 20, 30 and 60 hours, respectively. Pipes A, B and C are opened at 7 a.m., 8 a.m., & 9 a.m., respectively, on the same day. When will the tank be fulfilled?
SSC CGL TIER-II 03/02/2022
(a) 4:40 p.m. (b) 5:40 p.m.
(c) 6:20 p.m. (d) 7:20 p.m.
- Pipes A, B and C can fill a tank in 15, 30 and 40 hours, respectively. Pipes A, B and C are opened at 6 a.m., 8 a.m. & 10 a.m., respectively, on the same day. When will the tank be fulfilled?
SSC CPO 24/11/2020 (Shift-02)
(a) 3:20 p.m. (b) 11:20 p.m.
(c) 7:20 p.m. (d) 5:20 p.m.
- Pipes A, B and C can fill a tank in 30 h, 40 h and 60 h respectively. Pipes A, B and C are opened at 7 a.m., 8 a.m. and 10 a.m. respectively on the same day. When will the tank be fulfilled?
SSC CGL TIER-II 11/09/2019
(a) 10:00 p.m. (b) 10:20 p.m.
(c) 9:20 p.m. (d) 9:40 p.m.
- Pipe A and pipe B running together can fill a cistern in 6 minutes. If B takes 5 minutes more than A to fill it, then the time in which A and B will fill that cistern separately will be, respectively, _____
SSC CGL 26/07/2023 (Shift-02)
(a) 15 min and 10 min
(b) 15 min and 20 min
(c) 25 min and 20 min
(d) 10 min and 15 min
- A tank is filled in 4 hours by three pipes A, B and C. The pipe C is $1\frac{1}{2}$ times as fast as B and B is 3 times as fast as A. How many hours will pipe A alone take to fill the tank?
SSC CGL 24/08/2021 (Shift-03)
(a) 17 (b) 34
(c) 30 (d) 15
- Fill pipe P is 21 times faster than fill pipe Q. If Q can fill a cistern in 110 minutes, find the time it takes to fill the cistern when both fill pipes are opened together.
SSC CGL 09/09/2024 (Shift-01)
(a) 5 minutes (b) 4 minutes
(c) 3 minutes (d) 6 minutes
- One pipe can fill a tank 6 times faster than another pipe. If both the pipes together can fill the tank in 40 minutes, then in how many minutes will the slower pipe alone be able to fill the tank?
SSC CGL 24/09/2024 (Shift-01)
(a) 275 (b) 285
(c) 290 (d) 280

20. There are two pipes to fill a tank. Together, they can fill the tank in 15 minutes. If one pipe can fill the tank in one and a half times as fast as the other, the faster pipe alone can fill the tank in:
SSC CGL 11/09/2024 (Shift-01)
 (a) 10 minutes
 (b) 25 minutes
 (c) 20 minutes
 (d) $32\frac{1}{2}$ minutes
21. One pipe can fill a tank four times as fast as another pipe. If together the two pipes can fill the tank in 48 minutes, the slower pipe alone will be able to fill the tank in:
SSC CGL 26/09/2024 (Shift-01)
 (a) 192 min. (b) 288 min.
 (c) 240 min. (d) 144 min.
22. A tank is filled in 45 minutes by two pipes, A and B. Pipe B fills the tank twice as fast as A. How much time (in minutes) will pipe A alone take to fill the tank?
SSC CGL 26/09/2024 (Shift-03)
 (a) 135 (b) 140
 (c) 125 (d) 115
23. There are two pipes used to fill a tank and when operated together, they can fill the tank in 20 minutes. If one pipe can fill the tank two and a half time as quickly as the other, then the faster pipe alone can fill the tank in:
SSC CGL 13/09/2024 (Shift-01)
 (a) 30 min (b) 32 min
 (c) 34 min (d) 28 min
- ♦ TYPE - II ♦**
FILLING & EMPTYING SIMULTANEOUSLY
24. A tap can fill a cistern in 10 minutes and another tap can empty it in 12 minutes. If both the taps are open, the time (in hours) taken to fill the tank will be:
SSC CGL 25/09/2024 (Shift-03)
 (a) 1.5 hours (b) 2.5 hours
 (c) 2 hours (d) 1 hour
25. An inlet pipe can fill a water storage tank in 11 hours and an outlet pipe can empty the completely filled tank in 15 hours. If both pipes opened simultaneously. The time taken to fill the empty tank (in hrs) is :
SSC CGL 25/09/2024 (Shift-01)
 (a) $45\frac{1}{2}$ (b) $41\frac{1}{4}$
 (c) $49\frac{3}{4}$ (d) 40
26. An inlet can fill an empty tank in 51 hours while an outlet pipe drains a completely-filled tank in 76.5 hours. If both the pipes are opened simultaneously when the tank is empty, in how many hours will the tank get completely filled?
SSC CHSL 08/08/2023 (Shift-01)
 (a) 178.5 (b) 127.5
 (c) 102 (d) 153
27. An inlet pipe can fill an empty tank in 3.6 hours, while an outlet pipe can drain a completely-filled tank in 6.3 hours. If both the pipes are opened simultaneously when the tank is empty, in how many hours will the tank get completely filled?
SSC CPO 04/10/2023 (Shift-01)
 (a) 8.7 (b) 8.1
 (c) 9.0 (d) 8.4
28. P can fill a tank in 5 hours. Q can fill the same tank in 10 hours. R can empty the same tank in 20 hours. How much time will all the three take together to fill the tank?
SSC CHSL 13/03/2023 (Shift-02)
 (a) 10 hours (b) 4 hours
 (c) 6 hours (d) 5 hours
29. Two pipes can fill a tank in 15 hours and 4 hours, respectively, while a third pipe can empty it in 12 hours. How long (in hours) will it take to fill the empty tank if all the three pipes are opened simultaneously?
SSC CPO 23/11/2020 (Shift-02)
 (a) $\frac{20}{7}$ (b) $\frac{15}{7}$
 (c) $\frac{50}{7}$ (d) $\frac{30}{7}$
30. Two pipes can fill a tank in 12 hours and 18 hours, respectively, while a third pipe can empty it in 8 hours. How long (in hours) will it take to fill the empty tank if all three pipes are opened simultaneously?
SSC CGL 17/09/2024 (Shift-03)
 (a) 48 (b) 24
 (c) 36 (d) 72
31. Pipe A can fill a tank in 12 minutes; pipe B can fill it in 18 minutes, while pipe C can empty the full tank in 36 minutes. If all the pipes are opened simultaneously, how much time will it take to fill the empty tank completely?
SSC CGL 20/07/2023 (Shift-04)
 (a) 7 min. 30 sec. (b) 10 min.
 (c) 9 min. (d) 6 min.
32. Pipes A and B can empty a full tank in 16 hours and 24 hours, respectively. Pipe C alone can fill the empty tank in 4 hours. If A, B and C are opened together, then 35% tank will be filled in.
SSC GD 17/11/2021 (Shift-03)
 (a) $2\frac{1}{5}$ hours (b) $2\frac{2}{5}$ hours
 (c) $3\frac{1}{5}$ hours (d) $3\frac{2}{5}$ hours
33. Pipe A and B can fill a tank in 12 minutes and 15 minutes, respectively. The tank when full can be emptied by pipe C in x minutes. When all the three pipes are opened simultaneously, the tank is full in 10 minutes. The value of x is :
SSC CGL TIER-II 16/11/2020
 (a) 18 (b) 15
 (c) 20 (d) 24
34. Pipes P and Q can completely fill a water tank in 10 hours and 15 hours, respectively. A pipe R can empty a tank filled completely with water in 12 hours. Initially, the tank is empty and only pipes P and Q are opened at 6 a.m. and pipe R is also opened at 9 a.m. By what time will the tank be completely filled?
SSC CGL 10/09/2024 (Shift-03)
 (a) 1 p.m. (b) 2 p.m.
 (c) 11 p.m. (d) 3 p.m.
35. Pipe A can fill a cistern in 4 hours and Pipe B can fill the same cistern in 5 hours. Pipe C can empty a full cistern in 3 hours. If all three pipes are opened together, then the time (in minutes) taken to fill the tank is: (round to the nearest minute)
SSC CGL 24/09/2024 (Shift-03)
 (a) 625 (b) 445
 (c) 800 (d) 514
36. An inlet pipe can fill an empty tank in 140 hours while an outlet pipe drains a completely-filled tank in 63 hours. If 8 inlet pipes and y outlet pipes are opened simultaneously, when the tank is empty, then the tank gets completely filled in 105 hours. Find the value of y.
SSC CGL 14/07/2023 (Shift-03)
 (a) 1 (b) 4
 (c) 3 (d) 2
37. Inlet Pipes A and B can together fill an empty tank in 1.5 hours. Outlet Pipe C, when opened alone, can empty the completely filled tank, in 4.5 hours. When only Pipes A and C are opened together, the empty tank is filled in 6 hours. Find the time taken by Pipe B, when opened alone, to fill the empty tank.
SSC CGL 19/07/2023 (Shift-02)
 (a) 3 hours 30 minutes
 (b) 3 hours 36 minutes
 (c) 3 hours 32 minutes
 (d) 3 hours 40 minutes
38. An inlet pipe can fill an empty tank in 120 hours while an outlet pipe drains a completely-filled tank in 54 hours. If 8 inlet pipes and 3 outlet pipes are opened simultaneously, when the tank is empty, then in how many hours will the tank get completely filled?
SSC CGL 20/07/2023 (Shift-01)
 (a) 81 (b) 96
 (c) 72 (d) 90
39. Two pipes S_1 and S_2 alone can fill an empty tank in 15 hours and 20 hours respectively. Pipe S_3 alone can empty that completely filled tank in 40 hours. Firstly both pipes S_1 and S_2 are opened and after 2 hours pipe S_3 is also opened. In how much time tank will be completely filled after S_3 is opened?
SSC CGL 21/07/2023 (Shift-02)
 (a) $\frac{90}{17}$ hours (b) $\frac{89}{12}$ hours
 (c) $\frac{90}{13}$ hours (d) $\frac{92}{11}$ hours

40. Each inlet pipe can fill an empty cistern in 84 hours while each drain pipe can empty the same cistern from a filled condition in 105 hours. When the cistern is empty, 9 inlet pipes and 10 outlet pipes are simultaneously opened. After how many hours will the cistern be completely filled?

SSC CGL 26/07/2023 (Shift-02)

- (a) 84 (b) 88
(c) 80 (d) 90

41. There are two water taps in a tank which can fill the empty tank in 12 hours and 18 hours respectively. It is seen that there is a leakage point at the bottom of the tank which can empty the completely filled tank in 36 hours. If both the water taps are opened at the same time to fill the empty tank and the leakage point was repaired after 1 hour, then in how much time the empty tank will be completely filled?

SSC CGL 17/08/2021 (Shift-02)

- (a) 7 hours 12 minutes
(b) 8 hours 24 minutes
(c) 7 hours
(d) 7 hours 24 minutes

42. Pipes A, B and C can fill an empty tank in $\frac{30}{7}$ hours, if all the three pipes are opened simultaneously. A and B are filling pipes and C is an emptying pipe. Pipe A can fill the tank in 15 hours and pipe C can empty it in 12 hours. In how long (in hours) can pipe B alone fill the empty tank?

SSC CPO 24/11/2020 (Shift-01)

- (a) 4 (b) 6
(c) 3 (d) 5

43. Pipe A can fill a tank in 10 hours. Pipe B can fill the same tank in 12 hours. Pipe C can empty a full tank in 16 hours. All the pipes are opened at 8:00 a.m. and pipe A and B are closed at 10:00 a.m. After how much time from starting will the tank be empty?

SSC CGL 06/06/2019 (Shift-01)

- (a) 5 hours 52 minutes
(b) 5 hours 24 minutes
(c) 4 hours 30 minutes
(d) 4 hours 8 minutes

44. Pipe Q can fill the tank in 60 hours while pipe R may fill in 45 hours. Q and R pipes are opened together for 6 hours after which pipe W is also opened to empty the tank. All three pipes are opened simultaneously for 24 hours to reach the half level mark. How much time (in hours) will pipe W alone take to empty the entire tank?

SSC CGL 18/09/2024 (Shift-03)

- (a) 48 (b) 42
(c) 36 (d) 30

♦ TYPE - III ♦
TAPS OPENED ALTERNATELY

45. Pipe X can fill a tank in 9 hours and Pipe Y can fill it in 21 hours. If they are opened on alternate hours and Pipe X is opened first, in how many hours shall the tank be full?

SSC CGL 19/09/2024 (Shift-01)

- (a) $10\frac{3}{7}$ (b) $12\frac{3}{7}$
(c) $9\frac{3}{7}$ (d) $11\frac{3}{7}$

46. Two pipes A and B can fill an empty tank in 10 hours and 16 hours respectively. They are opened alternately for 1 hour each, opening pipe B first. In how many hours, will the empty tank be filled?

SSC CGL 20/08/2021 (Shift-02)

- (a) $12\frac{2}{5}$ (b) $14\frac{2}{5}$
(c) $10\frac{2}{5}$ (d) $16\frac{2}{5}$

47. An inlet pipe can fill an empty tank in $4\frac{1}{2}$ hours while an outlet pipe drains a completely filled tank in $7\frac{1}{5}$ hours. The tank is initially empty, and the two pipes are alternately opened for an hour each, till the tank is completely filled, starting with the inlet pipe. In how many hours will the tank be completely filled?

SSC CGL 21/07/2023 (Shift-02)

- (a) 24 (b) $20\frac{1}{4}$
(c) $20\frac{3}{4}$ (d) $22\frac{3}{8}$

48. A tank when full can be emptied by an outlet pipe A in 5.6 hours, while an inlet pipe B can fill the same empty tank in 7 hours. If pipes A and B are turned on alternatively for 1 hour each starting with pipe A when the tank is full, how long will it take to empty the tank?

SSC CGL TIER-II (20/01/2025)

- (a) 48 hours (b) 47 hours
(c) 56 hours (d) 55 hours

♦ TYPE - IV ♦
TO FIND THE CAPACITY OF TANK

49. There is a leak in a tank which empties it in 6 hours. A tap is turned on which fills the tank with 10 liters of water per minute. When it is full, it now takes 10 hours to empty. What is the capacity (in litres) of the tank?

SSC GD 18/11/2021 (Shift-02)

- (a) 8000 (b) 8500
(c) 9000 (d) 10000

50. Pipe A and B fill a tank in 43.2 minutes and 108 minutes respectively. Pipe C can empty it at 3 litres/minutes. When all the three pipes are opened together, they will fill the tank in 54 minutes. The capacity (in litres) of the tank is:

SSC CGL TIER-II 15/11/2020

- (a) 160 (b) 180
(c) 216 (d) 200

51. Two pipes can fill a tank separately in 36 minutes and 45 minutes respectively. A drain pipe installed in the tank can remove 40 litres of water per minute. If all three pipes are opened together, the tank is filled in one hour. Find the capacity/holding (in litres) of the tank.

SSC CHSL TIER-II 10/01/2024

- (a) 600 (b) 400
(c) 300 (d) 1200

52. At a school building, there is an overhead tank. To fill this tank 50 buckets of water are required. Assume that the capacity of the bucket is reduced to two-fifth of the present. How many buckets of water are required to fill the same tank?

SSC CGL 19/09/2024 (Shift-02)

- (a) 62.5 (b) 20
(c) 125 (d) 60

53. Pipe X can fill a tank in 60 hours while pipe Y can fill the tank in 72 hours. Both pipes are opened together for 20 hours. How much of the tank is left empty?

SSC CGL 18/09/2024 (Shift-01)

- (a) $\frac{1}{8}$ (b) $\frac{243}{360}$
(c) $\frac{7}{18}$ (d) $\frac{3}{8}$

♦ TYPE - V ♦
LEAKAGE IN TANK

54. A pump can fill a tank with water in 1 hour. Because of a leak, it took $1\frac{1}{3}$ hours to fill the tank. In how many hours can the leak alone drain all the water of the tank when it is full?

SSC CPO 25/11/2020 (Shift-02)

- (a) 2 (b) 1
(c) 4 (d) 5

55. A pump can fill a tank with water in 3 hours. Because of a leak, it took $3\frac{1}{3}$ hours to fill the tank. In how many hours can the leak alone drain all the water of the tank when it is full?

SSC CPO 24/11/2020 (Shift-02)

- (a) 21 (b) 15
(c) 30 (d) 10

56. A pipe can fill a tank in 30 hours. Due to a leakage at the bottom, it is filled in 50 hours. How much time will the leakage take to empty the completely filled tank?

SSC CGL 24/07/2023 (Shift-02)

- (a) 60 hours (b) 85 hours
(c) 70 hours (d) 75 hours

57. A pipe can fill an overhead tank in 12 hours. But due to a leak at the bottom, it is filled in 18 hours. If the tank is full, how much time will the leak take to empty it?

SSC CGL 09/09/2024 (Shift-03)

- (a) 3.6 hours (b) 63 hours
(c) 7.2 hours (d) 36 hours

58. Pipe A usually fills a tank in 6 hours. But due to a leak at the bottom of the tank, it takes extra 2 hours to fill the tank. If the tank is full, then how much time will it take to get emptied due to the leak?

SSC CGL 11/09/2024 (Shift-02)

- (a) 16 hours (b) 20 hours
(c) 12 hours (d) 24 hours

59. A tap can fill a tank in $5\frac{1}{2}$ hours.

Because of a leak, it took $8\frac{1}{4}$ hours to fill the tank. In how much time (in hours) will the leak alone empty 30% of the tank?

SSC CGL TIER-II 29/01/2022

- (a) $\frac{99}{20}$ (b) $\frac{5}{2}$
(c) $\frac{9}{2}$ (d) $\frac{17}{2}$

60. Two pipes A and B can fill a cistern in $12\frac{1}{2}$ hours and 25 hours, respectively. The pipes were opened simultaneously, and it was found that, due to leakage in the bottom, it took one hour 40 minutes more to fill the cistern. If the cistern is full, in how much time (in hours) will the leak alone empty 70% of the cistern?

SSC CGL TIER-II 29/01/2022

- (a) 35 (b) 40
(c) 30 (d) 50

61. Two pipes A and B can fill a tank in 20 and 30 hours, respectively. Both pipes are opened to fill the tank, but when the tank is one-third full, a leak develops through which one-fourth of the water supplied by both pipes goes out. Find the total time (in hours) taken to fill the tank.

SSC CGL 23/09/2024 (Shift-03)

- (a) $14\frac{2}{3}$ (b) 14
(c) $11\frac{2}{5}$ (d) $12\frac{1}{3}$

62. Pipe A and B can fill a tank in 10 hours and 40 hours, respectively. C is an outlet pipe attached to the tank. If all the three pipes are opened simultaneously, it takes 80 minutes more time than what A and B together take to fill the tank. A and B are kept open for 7 hours and then closed and C is opened. C will now empty the tank in:

SSC CGL 06/06/2019 (Shift-01)

- (a) 45.5 hours (b) 38.5 hours
(c) 42 hours (d) 49 hours

63. Two pipes can fill a cistern in 12 hours and 16 hours, respectively. The pipes are opened simultaneously and it is found that due to leakage at the bottom, it takes 90 minutes more to fill the cistern. How much time will the leakage take to empty the completely filled tank?

SSC CPO 04/10/2023 (Shift-03)

- (a) $39\frac{13}{49}$ h (b) $36\frac{29}{49}$ h
(c) $37\frac{15}{49}$ h (d) $38\frac{10}{49}$ h

♦ TYPE - VI ♦
WHEN 1 OR 2 TAPS ARE
CLOSED DURING THE FILLING
OR EMPTYING A TANK

64. Two taps P and Q can fill a tank alone in 10 hours and 12 hours respectively. If the two taps are opened at 9 a.m., then at what time should the tap P be closed to completely fill the tank at exactly 3 p.m.?

SSC MTS 16/06/2023 (Shift-01)

- (a) 2 p.m (b) 1 p.m
(c) 3 p.m (d) 12 p.m

65. Pipes A and B can fill a tank in 18 hours and 27 hours, respectively, whereas pipe C can empty the full tank in 54 hours. All three pipes are opened together, but pipe C is closed after 12 hours. In how much time (in minutes) will the one-third of the remaining part of the tank be filled by A and B together?

SSC Phase XII 21/06/2024 (Shift-02)

- (a) 24 (b) 36
(c) 30 (d) 15

66. Pipes A, B and C can fill a tank in 30 hours, 36 hours and 28 hours, respectively. All the three pipes were opened simultaneously. If A and C were closed 5 hours and 8 hours, respectively, before the tank was filled completely, then in how many hours was the tank filled?

SSC MTS 08/10/2021 (Shift-03)

- (a) 14 (b) 15
(c) 12 (d) 16

67. Pipes A and B can fill a tank in 12 hours and 16 hours respectively and pipe C can empty the full tank in 24 hours. All three pipes are opened together, but after 4 hours pipe B is closed. In how many hours, the empty tank will be completely filled?

SSC CGL 23/08/2021 (Shift-01)

- (a) 18 (b) 32
(c) 28 (d) 14

68. Pipe A and B can fill a tank in 16 hours and 24 hours respectively, and pipe C alone can empty the full tank in x hours. All the pipes were opened together at 10.30 a.m., but C was closed at 2.30 p.m. If the tank was full at 8.30 p.m. on the same day, then what is the value of x?

SSC CGL TIER-II 12/09/2019

- (a) 64 (b) 48
(c) 45 (d) 96

69. Pipes M, N and S can fill a tank in 25, 50 and 100 minutes, respectively. Initially, pipes N and S are kept open for 10 minutes, and then pipe N is shut while pipe M is opened. Pipe S is closed 15 minutes before the tank overflows. How much time (in minutes) will it take to fill the tank if the three pipes work in this pattern?

SSC CGL 09/09/2024 (Shift-02)

- (a) 30 (b) 33
(c) 42 (d) 27

70. Three pipes P, Q and R can fill a cistern in 40 minutes, 80 minutes and 120 minutes, respectively. Initially, all the pipes are opened. After how much time (in minutes) should the pipes Q and R be turned off so that the cistern will be completely filled in just half an hour?

SSC CGL 10/09/2024 (Shift-01)

- (a) 14 (b) 10
(c) 16 (d) 12

71. Two pipes, A and B, can fill a tank in 10 minutes and 20 minutes, respectively. The pipe C can empty the tank in 30 minutes. All the three pipes are opened at a time in the beginning. However, pipe C is closed 2 minutes before the tank is filled. In what time, will the tank be full (in minutes)?

SSC CGL 12/09/2024 (Shift-01)

- (a) 12 (b) 10
(c) 8 (d) 6

72. Two pipes A and B can fill a tank in 48 minutes and 66 minutes, respectively. If both the pipes are opened simultaneously, then after how many minutes should pipe B be closed so that the tank gets filled in 32 minutes?

SSC CGL 12/09/2024 (Shift-02)

- (a) 18 (b) 16
(c) 22 (d) 20

73. Pipes A and B can fill a tank in 15 hours and 25 hours, respectively, whereas pipe C can empty the full tank in 40 hours. All three pipes are opened together, but pipe A is closed after 5 hours. After how many hours will the remaining part of the tank be filled?

SSC CGL 17/09/2024 (Shift-01)

- (a) $41\frac{4}{9}$ (b) $43\frac{4}{9}$
(c) $44\frac{4}{9}$ (d) $39\frac{4}{9}$

74. Pipes A and B can fill a tank in 6 hours and 8 hours, respectively. Both pipes are opened together for 3 hours. After that pipe A is closed, and B continues to fill the tank. In how many hours will the tank be filled?

SSC CGL 18/09/2024 (Shift-02)

- (a) 2 (b) 4
(c) 3 (d) 6

Chapter - 13 : Pipe and Cistern

75. Two inlet pipes, P_1 and P_2 , can fill a cistern in 20 hours and 30 hours, respectively. They were opened at the same time, but pipe P_1 had to be closed 5 hours before the cistern was full. How many hours in total did it take for the two pipes to fill the cistern?
SSC CGL 19/09/2024 (Shift-03)
(a) 15 hrs. (b) 9 hrs.
(c) 12 hrs. (d) 10 hrs.
76. Three pipes, P, Q and R, together take four hours to fill a tank. All the three pipes were opened at the same time. After three hours, P was closed, and Q and R filled the remaining tank in two hours. How many hours will P alone take to fill the tank?
SSC CGL 23/09/2024 (Shift-02)
(a) 8 hours (b) 10 hours
(c) 12 hours (d) 9 hours
77. There are 3 taps A, B and C in a tank. These can fill the tank in 10 hours, 20 hours and 25 hours, respectively. At first, all three taps are opened simultaneously. After 2 hours, tap C is closed and tap A and hours B keep running. After 4 hours, tap B is also closed. The remaining tank is filled by tap A alone. Find the percentage of work done by tap A itself.
SSC CPO 10/11/2022 (Shift-02)
(a) 75% (b) 52%
(c) 72% (d) 32%
78. Two pipes A and B can fill a tank in 12 minutes and 24 minutes, respectively, while a third pipe C can empty the full tank in 32 minutes. All the three pipes are opened simultaneously. However, pipe C is closed 2 minutes before the tank is filled. In how much time (in minutes) will the tank be full?
SSC CGL TIER-II 03/02/2022
(a) 9 (b) 10
(c) 12 (d) 8
79. Pipes A and B can empty a full tank in 36 minutes and 45 minutes, respectively, whereas pipe C alone can fill the tank in 15 minutes. B and C are opened together for 15 minutes, and then both are turned off and A is opened. Pipe A will empty the tank (in minutes);
SSC CHSL 19/04/2021 (Shift-02)
(a) 20 (b) 24
(c) 30 (d) 18
80. Pipes A and B can fill a tank in 16 hours and 24 hours, respectively, whereas pipe C can empty the full tank in 40 hours. All three pipes are opened together, but pipe A is closed after 10 hours. After how many hours will the remaining part of the tank be filled?
SSC CPO 23/11/2020 (Shift-01)
(a) $15\frac{1}{2}$ (b) $12\frac{1}{2}$
(c) 20 (d) 10
81. Pipes A and B can fill a tank in 16 hours and 24 hours, respectively, whereas pipe C can empty the full tank in 40 hours. All three pipes are opened together, but pipe C is closed after 10 hours. After how many hours will the remaining part of the tank be filled?
SSC CPO 25/11/2020 (Shift-01)
(a) 2 (b) $2\frac{1}{2}$
(c) 5 (d) $5\frac{1}{2}$
82. Pipe A and B are filling pipes while C is an emptying pipe. A and B can fill a tank in 72 minutes and 90 minutes respectively. When all the three pipes are opened together, the tank gets filled in 2 hours. A and B are opened together for 12 minutes, then closed and C is opened. The tank will be empty after
SSC CGL TIER-II 13/09/2019
(a) 15 minutes (b) 18 minutes
(c) 12 minutes (d) 16 minutes
83. Pipes A and B can fill a tank in one hour and two hours respectively while pipe C can empty the filled up tank in one hour and fifteen minutes. A and C are turned on together at 9 a.m. After 2 hours, only A is closed and B is turned on. When will the tank be emptied?
SSC CGL 06/06/2019 (Shift-01)
(a) 12:10 p.m.
(b) 11:30 a.m.
(c) 10:30 a.m.
(d) 12:20 p.m.
84. A cistern can be filled by two pipes in 8 minutes and 10 minutes, separately. Both the pipes are opened together for a certain time, but due to an obstruction, only $\frac{5}{8}$ of the full quantity of water flowed through the former pipe and $\frac{3}{5}$ through the latter pipe. However, the obstruction was suddenly removed, and the cistern was filled in 3 minutes from that moment. How long did it take before the full flow began?
SSC CGL 17/09/2024 (Shift-02)
(a) $3\frac{1}{16}$ minutes (b) $2\frac{6}{17}$ minutes
(c) $9\frac{6}{7}$ minutes (d) $2\frac{30}{37}$ minutes
85. Pipe A takes $\frac{4}{5}$ of the time required by pipe B to fill an empty tank individually. When an outlet pipe C is also opened simultaneously with pipes A and B, it takes $\frac{4}{5}$ more time to fill the empty tank than it takes when only pipe A and pipe B are opened together. If it takes 40 hours to fill the tank when all the three pipes are opened simultaneously, in what time (in hours) will pipe C empty the full tank, operated alone?
SSC CGL 24/09/2024 (Shift-02)
(a) 50 hours (b) 45 hours
(c) 65 hours (d) 75 hours
86. Pipe A and B can fill a tank in 10 hours and 40 hours, respectively. C is an outlet pipe attached to the tank. If all the three pipes are opened simultaneously, it takes 80 minutes more time than what A and B together take to fill the tank. A and B are kept open for 7 hours and then closed and C is opened. C will now empty the tank in :
SSC CGL 06/06/2019 (Shift-01)
(a) 45.5 hours
(b) 38.5 hours
(c) 42 hours
(d) 49 hours
87. When operated separately, pipe A takes 5 hours less than pipe B to fill a cistern and when both pipe are operated together, the cistern get filled in 6 hours. In how much time (in hours) will pipe B fill the cistern, if operated?
SSC CPO 23/11/2020 (Shift-03)
(a) 9
(b) 18
(c) 10
(d) 15
88. The tank is filled by three pipes with different uniform flow rates. While the first two pipes are operating simultaneously, they fill the tank in the same duration that the third pipe takes to fill it alone. The second pipe can fill the tank 5 hours quicker than the first pipe, yet 4 hours slower than the third pipe. What is the time (in hours) needed for the first pipe to fill the tank?
SSC CGL 11/09/2024 (Shift-03)
(a) 18 (b) 12
(c) 9 (d) 15



ANSWER KEY

1.	(d)	2.	(b)	3.	(d)	4.	(c)	5.	(a)	6.	(c)	7.	(c)	8.	(c)	9.	(c)	10.	(b)
11.	(b)	12.	(c)	13.	(b)	14.	(a)	15.	(c)	16.	(d)	17.	(b)	18.	(a)	19.	(d)	20.	(b)
21.	(c)	22.	(a)	23.	(d)	24.	(d)	25.	(b)	26.	(d)	27.	(d)	28.	(b)	29.	(d)	30.	(d)
31.	(c)	32.	(b)	33.	(c)	34.	(d)	35.	(d)	36.	(c)	37.	(b)	38.	(d)	39.	(d)	40.	(a)
41.	(d)	42.	(a)	43.	(a)	44.	(c)	45.	(b)	46.	(a)	47.	(c)	48.	(b)	49.	(c)	50.	(c)
51.	(d)	52.	(c)	53.	(c)	54.	(c)	55.	(c)	56.	(d)	57.	(d)	58.	(d)	59.	(a)	60.	(a)
61.	(a)	62.	(d)	63.	(d)	64.	(a)	65.	(a)	66.	(b)	67.	(a)	68.	(d)	69.	(d)	70.	(d)
71.	(c)	72.	(c)	73.	(d)	74.	(b)	75.	(a)	76.	(a)	77.	(c)	78.	(b)	79.	(b)	80.	(b)
81.	(a)	82.	(b)	83.	(d)	84.	(b)	85.	(a)	86.	(d)	87.	(d)	88.	(d)				

SOLUTIONS

1. (d) A → 15 hrs $\frac{6}{5}$
B → 18 hrs $\frac{5}{6}$
Time taken by A and B to fill the empty tank = $\frac{90}{11} = 8\frac{2}{11}$ hours

Alternate Method:

Time taken to fill the empty tank
= $\frac{15 \times 18}{15 + 18} = \frac{15 \times 18}{33} = 8\frac{2}{11}$ hrs

2. (b) P → 36 min. $\frac{5}{18}$ Total capacity
Q → 45 min. $\frac{4}{9}$
Time taken by both pipes to fill the tank = $\frac{180}{5+4} = 20$ min.

3. (d) A → 12 hours $\frac{3}{4}$
B → 18 hours $\frac{2}{3}$
Total time taken = $\frac{36}{3+2} = \frac{36}{5}$
= 7 hrs 12 min.

Alternate Method:-

Total time taken = $\frac{12 \times 18}{12 + 18}$
= $\frac{12 \times 18}{30} = 7$ hr 12 min.

4. (c) L → 30h $\frac{3}{10}$ Eff. Capacity
M → 45h $\frac{2}{9}$
Time taken by both pipe to fill
= $\frac{90}{3+2} = 18$ hrs.

5. (a) X → 14 hours $\frac{3}{7}$ Eff. Total capacity
Y → 21 hours $\frac{2}{3}$
Time required to fill tank
= $\frac{42}{3+2} = \frac{42}{5} = 8\frac{2}{5}$ hours

Alternate Method:

$$\frac{14 \times 21}{14 + 21} = \frac{42}{5} = 8\frac{2}{5} \text{ hours}$$

6. (c) A → 18 h $\frac{4}{9}$
B → 24 h $\frac{3}{8}$
Time taken by A and B to fill the tank
= $\frac{72}{4+3} = \frac{72}{7} = 10\frac{2}{7}$ h

Alternate Method:

$$\frac{18 \times 24}{18 + 24} = \frac{72}{7} = 10\frac{2}{7} \text{ h}$$

7. (c) A → $\frac{4}{3}$ hrs. $\frac{3}{4}$ Eff. Total capacity
B → 2 hrs. $\frac{1}{2}$
Time taken by both pipes to fill the tank = $\frac{4}{3+2} = \frac{4}{5}$ hr. = $\frac{4}{5} \times 60$ min. = 48 min.

8. (c) A → 12 hrs $\frac{3}{4}$
B → 18 hrs $\frac{2}{3}$
Time taken by pipe A & B to fill the tank = $\frac{36}{5} \times 60 = 432$ min

9. (c) $\frac{M_1 H_1 D_1}{W_1} = \frac{M_2 H_2 D_2}{W_2}$
ATQ,
 $5 \times \frac{5}{3} = 4 \times T_2$
 $\Rightarrow T_2 = \frac{25}{12}$ hours = $\frac{25}{12} \times 60$ minutes = 125 minutes

Alternate Method:

$$\text{Efficiency} \propto \frac{1}{\text{time}}$$

$$\begin{array}{l} 5 : 4 \\ 4 : 5 \\ 100 : 125 \text{ min} \end{array}$$

10. (b) Fill a tank capacity = (15 × 36 min)
To fill the tank in 1 hour
Required Tap = $\frac{15 \times 36}{60} = 9$

11. (b) A → 5 min $\frac{6}{5}$ Eff. Total capacity
B → 6 min $\frac{5}{6}$
∴ Required time = $\frac{30}{6-5} = 30$ mins

12. (c) A → 4 hrs $\frac{3}{4}$ Eff. Total capacity
B → 6 hrs $\frac{2}{3}$
Let capacity of tank be 12 units
ATQ,
A + x(A + B) = 12
 $\Rightarrow 3 + 5x = 12 \Rightarrow x = \frac{9}{5}$ hrs
∴ Total time = $\frac{9}{5} + 1 = \frac{14}{5}$ hrs.

$$= 2 \text{ hrs} + \frac{4}{5} \text{ hrs} = 2 \text{ hrs } 48 \text{ min}$$

Time = 8 am. + 2 hrs 48 min

= 10 : 48 am

Alternate Method:

A → 4 hrs. $\frac{3}{4}$ Eff. Total capacity
B → 6 hrs. $\frac{2}{3}$
Tank filled by A in 1 hr. = 3 units
Remaining part = 12 - 3 = 9 units
Time taken by A and B to fill remaining part = $\frac{9}{5} = 1\frac{4}{5}$ hrs.
= 1 hr. 48 min.
∴ Time = 9 am. + 1 hr. 48 min.
= 10 : 48 am.

13. (b) 7AM A → 20 hrs $\frac{3}{4}$
8AM B → 30 hrs $\frac{2}{3}$
9AM C → 60 hrs $\frac{1}{2}$
By A till 9 AM = 2 × 3 = 6 units
By B till 9 AM = 2 × 1 = 2 units
Remaining work = 60 - (6 + 2) = 52 units
Required time = $\frac{52}{6} = 8\frac{2}{3}$ hrs
= 8 hrs 40 min
Time = 9 AM + 8 hrs 40 min = 5 : 40 PM